Wireless Receiver Design: Gain, Noise Figure, and Linearity Specifications

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***Abstract*—This report documents the system-level design of a wireless receiver, focusing on gain, noise figure, and linearity specifications. It presents the methodology for assigning specifications to the building blocks of a ZigBee receiver and includes Matlab code used for performance calculations.**

I. Introduction

Wireless communication standards such as ZigBee (IEEE 802.15.4) specify stringent requirements for receiver sensitivity, bit-error rate (BER), interference rejection, and more. The system-level design process maps these standards onto electrical specifications for receiver components [1].

II. Problem Statement

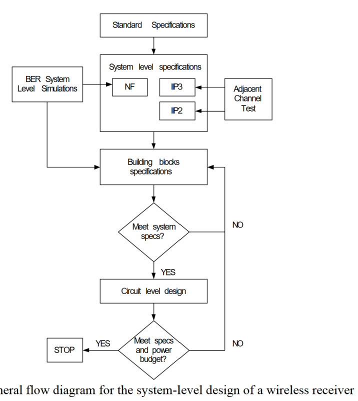


Fig. 1: General flow diagram for the system-level design of a wireless receiver

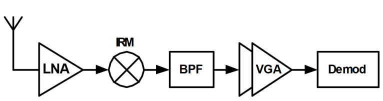


Fig. 2: Figure 2. Low-IF Architecture for a ZigBee Receiver

The objective is to obtain the gain, noise figure (NF), and linearity (IIP3) specifications for the receiver building blocks to achieve a given overall performance. For the ZigBee standard, key requirements are listed in Table I.

TABLE I: ZigBee Standard Physical Layer Specifications

Carrier	2.4 GHz
Sensitivity	-85 dBm
BER	5×10^{-5}
Required SNR	2 dB
Channel bandwidth	2 MHz
Alternate channel rejection	30 dB

III. Calculations

A. Total Receiver Gain

$$\begin{aligned} G &= \text{Required signal level at the input of demodulator} \\ &\quad - \text{Minimum detectable signal} \qquad (1) \\ &= 0\,\text{dBm} - (-85\,\text{dBm}) = 85\,\text{dB} \qquad (2) \end{aligned}$$

$$\begin{aligned} NF &= \text{Sensitivity} - (-174 + 10\log_{10}(BW)) \\ &\quad - \text{SNR} - NF_{\text{margin}} \qquad (3) \\ &= -85\,\text{dBm} - (-174 + 10\log_{10}(2\times10^6)) \\ &\quad - 2\,\text{dB} - 3\,\text{dB} \qquad (4) \\ &= 20\,\text{dB} \qquad (5) \end{aligned}$$

B. Total Receiver Noise Figure

$$\begin{aligned} NF &= \text{Sensitivity} - (-174 + 10\log_{10}(BW)) - \text{SNR} - NF_{\text{margin}} \qquad (6) \\ &= -85\,\text{dBm} - (-174 + 10\log_{10}(2\times10^6)) - 2\,\text{dB} - 3\,\text{dB} \qquad (7) \\ &= 20\,\text{dB} \qquad (8) \end{aligned}$$

C. Total Receiver IIP3

$$\text{Signal level (alternate channel test)}: -82\,\mathrm{dBm} \tag{9}$$

$$C2IR: 5\,\mathrm{dB} \tag{10}$$

$$\text{Alternate channel level}: -52\,\mathrm{dBm} \tag{11}$$

$$IM3: -87\,\mathrm{dBm} \tag{12}$$

$$IIP3 = \frac{-52-(-87)}{2} + (-52) + 3 \tag{13}$$

$$= -31.5\,\mathrm{dBm} \tag{14}$$

IV. Matlab Code

```
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% ECEN 665 { Laboratory 1
% Gain, Noise Figure and Linearity Analysis for a ZigBee Receiver
% Author: <Your Name>
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

clear; clc; close all;

%% ===========================
% GLOBAL CONSTANTS (Table 2)

BW = 2e6;                   % Channel bandwidth (Hz)
SNR_req = 2;                % Required SNR (dB)
NF_margin = 3;              % Noise figure margin (dB)
InsertionLoss = 2;          % Passive RF band-selection filter loss (dB)

kT = -174 + 10*log10(BW);   % Thermal noise over BW (dBm)

%% ==========================================================================
%% =========================== EXERCISE 1 ===============================
% ZigBee Standard Receiver (Sensitivity = -85 dBm)
% Includes passive RF band-selection filter

Sensitivity = -85;
Demod_input = 0;

blocks = {'RF Filter','LNA','IRM','BPF','VGA','Demod'};
Gain_dB = [-InsertionLoss, 14, 11, 0, 60, 0];
NF_dB   = [InsertionLoss, 4, 30, 40, 40, 0];

%% ---- Total Gain
TotalGain_Ex1 = Demod_input - Sensitivity;

%% ---- Required System NF (Section III equation)
NF_sys_Ex1 = Sensitivity - kT - SNR_req - NF_margin;

%% ---- Friis Noise Figure Accumulation
G = 10.^(Gain_dB/10);
NF = 10.^(NF_dB/10);

NF_cum = zeros(1,length(blocks));
NF_cum(1) = NF(1);

for k = 2:length(blocks)
    NF_cum(k) = NF_cum(k-1) + (NF(k)-1)/prod(G(1:k-1));
end

NF_cum_dB = 10*log10(NF_cum);

%% ---- Signal Levels (input of each block)
Signal = zeros(1,length(blocks));
Signal(1) = Sensitivity;

for k = 2:length(blocks)
    Signal(k) = Signal(k-1) + Gain_dB(k-1);
end

%% ---- Noise Levels (input of each block)
Noise = kT + NF_cum_dB;

%% ---- SNR Levels
SNR = Signal - Noise;

%% ---- IIP3 Calculation (Section III)
Signal_ACT = Sensitivity + 3;
C2IR = 5;
Alt_Channel = Signal_ACT + 30;
IM3 = Signal_ACT - C2IR;
IIP3_Ex1 = (Alt_Channel - IM3)/2 + Alt_Channel + 3;

%% ---- FIGURES (Exercise 1)

figure;
plot(Signal,'-o','LineWidth',1.5); grid on;
title('Exercise 1: Signal Levels at Input of Each Receiver Block');
ylabel('Signal Power (dBm)');
xticks(1:length(blocks)); xticklabels(blocks);

figure;
plot(Noise,'-o','LineWidth',1.5); grid on;
title('Exercise 1: Noise Levels at Input of Each Receiver Block');
ylabel('Noise Power (dBm)');
xticks(1:length(blocks)); xticklabels(blocks);

figure;
plot(SNR,'-o','LineWidth',1.5); grid on;
title('Exercise 1: SNR at Input of Each Receiver Block');
ylabel('SNR (dB)');
xticks(1:length(blocks)); xticklabels(blocks);


%% ===========================
% ================= EXERCISE 2 =================
% Improved Sensitivity Receiver (Sensitivity = -98 dBm)

Sensitivity = -98;

% ---- System-level specifications
TotalGain_Ex2 = Demod_input - Sensitivity;   % dB
NF_sys_Ex2 = Sensitivity - kT - SNR_req - NF_margin;  % dB

% ---- IIP3 Calculation (same methodology as Section III)
Signal_ACT = Sensitivity + 3;        % dBm
C2IR = 5;                            % dB
Alt_Channel = Signal_ACT + 30;       % dBm
IM3 = Signal_ACT - C2IR;             % dBm
IIP3_Ex2 = (Alt_Channel - IM3)/2 + Alt_Channel + 3;  % dBm

%% ---- DISPLAY RESULTS CLEARLY
disp('==================================================')
disp('EXERCISE 2: SYSTEM-LEVEL SPECIFICATIONS (Sensitivity = -98 dBm)')
disp('--------------------------------------------------')
fprintf('Required Total Receiver Gain     : %.1f dB\n', TotalGain_Ex2);
fprintf('Required Overall Noise Figure    : %.1f dB\n', NF_sys_Ex2);
fprintf('Required System IIP3             : %.1f dBm\n', IIP3_Ex2);
disp('==================================================')


%% =========================== EXERCISE 3 ===============================
% CMOS Receiver Design for Sensitivity = -98 dBm

blocks = {'RF Filter','LNA','IRM','BPF','VGA','Demod'};
Gain_dB = [-InsertionLoss, 18, 12, 0, 58, 0];
NF_dB   = [InsertionLoss, 3, 10, 15, 15, 0];

G = 10.^(Gain_dB/10);
NF = 10.^(NF_dB/10);

NF_cum = zeros(1,length(blocks));
NF_cum(1) = NF(1);

for k = 2:length(blocks)
    NF_cum(k) = NF_cum(k-1) + (NF(k)-1)/prod(G(1:k-1));
end

NF_cum_dB = 10*log10(NF_cum);

Signal(1) = Sensitivity;
for k = 2:length(blocks)
    Signal(k) = Signal(k-1) + Gain_dB(k-1);
end

Noise = kT + NF_cum_dB;
SNR = Signal - Noise;

%% ---- FIGURES (Exercise 3)

figure;
plot(Signal,'-o','LineWidth',1.5); grid on;
title('Exercise 3: Signal Levels at Input of Each Block (CMOS Receiver)');
ylabel('Signal Power (dBm)');
xticks(1:length(blocks)); xticklabels(blocks);

figure;
plot(Noise,'-o','LineWidth',1.5); grid on;
title('Exercise 3: Noise Levels at Input of Each Block (CMOS Receiver)');
ylabel('Noise Power (dBm)');
xticks(1:length(blocks)); xticklabels(blocks);

figure;
plot(SNR,'-o','LineWidth',1.5); grid on;
title('Exercise 3: SNR at Input of Each Block (CMOS Receiver)');
ylabel('SNR (dB)');
xticks(1:length(blocks)); xticklabels(blocks);

%% ==========================================================================
%% ==========================================================================
%% =========================== EXERCISE 4 ===============================
% Sheng et al. Power-Optimized System-Level Design
% Sensitivity = -98 dBm

Sensitivity = -98;

%% ===========================
% Exercise 3 (REFERENCE { CMOS BASELINE)
blocks3 = {'RF Filter','LNA','IRM','BPF','VGA','Demod'};
Gain3_dB = [-InsertionLoss, 18, 12, 0, 58, 0];
NF3_dB   = [InsertionLoss, 3, 10, 15, 15, 0];

%% ===========================
% Exercise 4 (Sheng et al. Optimized Allocation)
% More gain + lower NF early, relaxed baseband blocks

blocks4 = blocks3;
Gain4_dB = [-InsertionLoss, 22, 15, 0, 51, 0];   % Shift gain forward
NF4_dB   = [InsertionLoss, 2.5, 7, 18, 22, 0];   % Relax VGA/BPF NF

%% ---------- FUNCTION-LIKE CALCULATION ----------
% EXERCISE 3 CALCULATIONS
G3 = 10.^(Gain3_dB/10);
NF3 = 10.^(NF3_dB/10);
```

```
NF3_cum = zeros(1,length(blocks3));
NF3_cum(1) = NF3(1);
for k = 2:length(blocks3)
    NF3_cum(k) = NF3_cum(k-1) + (NF3(k)-1)/prod(G3(1:k-1));
end
NF3_cum_dB = 10*log10(NF3_cum);

Signal3 = zeros(1,length(blocks3));
Signal3(1) = Sensitivity;
for k = 2:length(blocks3)
    Signal3(k) = Signal3(k-1) + Gain3_dB(k-1);
end
Noise3 = kT + NF3_cum_dB;
SNR3 = Signal3 - Noise3;

% EXERCISE 4 CALCULATIONS
G4 = 10.^(Gain4_dB/10);
NF4 = 10.^(NF4_dB/10);

NF4_cum = zeros(1,length(blocks4));
NF4_cum(1) = NF4(1);
for k = 2:length(blocks4)
    NF4_cum(k) = NF4_cum(k-1) + (NF4(k)-1)/prod(G4(1:k-1));
end
NF4_cum_dB = 10*log10(NF4_cum);

Signal4 = zeros(1,length(blocks4));
Signal4(1) = Sensitivity;
for k = 2:length(blocks4)
    Signal4(k) = Signal4(k-1) + Gain4_dB(k-1);
end
Noise4 = kT + NF4_cum_dB;
SNR4 = Signal4 - Noise4;

%% ===========================
%% ======================= COMPARISON PLOTS ==============================

% -------- Signal Level Comparison
figure;
plot(Signal3,’-o’,’LineWidth’,1.5); hold on;
plot(Signal4,’-s’,’LineWidth’,1.5);
grid on;
title(’Exercise 3 vs Exercise 4: Signal Levels at Block Inputs’);
ylabel(’Signal Power (dBm)’);
legend(’Exercise 3: CMOS Baseline’,’Exercise 4: Sheng Optimized’,’Location’,’Best’);
xticks(1:length(blocks3)); xticklabels(blocks3);

% -------- Noise Level Comparison
figure;
plot(Noise3,’-o’,’LineWidth’,1.5); hold on;
plot(Noise4,’-s’,’LineWidth’,1.5);
grid on;
title(’Exercise 3 vs Exercise 4: Noise Levels at Block Inputs’);
ylabel(’Noise Power (dBm)’);
legend(’Exercise 3: CMOS Baseline’,’Exercise 4: Sheng Optimized’,’Location’,’Best’);
xticks(1:length(blocks3)); xticklabels(blocks3);

% -------- SNR Comparison
figure;
plot(SNR3,’-o’,’LineWidth’,1.5); hold on;
plot(SNR4,’-s’,’LineWidth’,1.5);
grid on;
title(’Exercise 3 vs Exercise 4: SNR at Block Inputs’);
ylabel(’SNR (dB)’);
legend(’Exercise 3: CMOS Baseline’,’Exercise 4: Sheng Optimized’,’Location’,’Best’);
xticks(1:length(blocks3)); xticklabels(blocks3);



%% =========================== EXERCISE 5 ================================
% IF Amplifier + ADC Architecture

blocks = {’RF Filter’,’LNA’,’IRM’,’IF Amp’,’BPF’,’VGA’,’ADC’};
Gain_dB = [-InsertionLoss, 18, 12, 14, 0, 50, 0];
NF_dB   = [InsertionLoss, 3, 10, 6, 15, 15, 0];

ADC_input = 8;
Sensitivity = -98;

TotalGain_Ex5 = ADC_input - Sensitivity;

G = 10.^(Gain_dB/10);
NF = 10.^(NF_dB/10);

NF_cum = zeros(1,length(blocks));
NF_cum(1) = NF(1);

for k = 2:length(blocks)
    NF_cum(k) = NF_cum(k-1) + (NF(k)-1)/prod(G(1:k-1));
end

NF_cum_dB = 10*log10(NF_cum);

Signal(1) = Sensitivity;
for k = 2:length(blocks)
    Signal(k) = Signal(k-1) + Gain_dB(k-1);
end

Noise = kT + NF_cum_dB;
SNR = Signal - Noise;

%% ---- FIGURES (Exercise 5)

figure;
plot(Signal,’-o’,’LineWidth’,1.5); grid on;
title(’Exercise 5: Signal Levels with IF Amplifier and ADC’);
ylabel(’Signal Power (dBm)’);
xticks(1:length(blocks)); xticklabels(blocks);

figure;
plot(Noise,’-o’,’LineWidth’,1.5); grid on;
title(’Exercise 5: Noise Levels with IF Amplifier and ADC’);
ylabel(’Noise Power (dBm)’);
xticks(1:length(blocks)); xticklabels(blocks);

figure;
plot(SNR,’-o’,’LineWidth’,1.5); grid on;
title(’Exercise 5: SNR with IF Amplifier and ADC’);
ylabel(’SNR (dB)’);
xticks(1:length(blocks)); xticklabels(blocks);
%%-----------------------------------------------------------------

Ex1_Table = table( ...
    blocks’, ...
    Gain_dB’, ...
    NF_cum_dB’, ...
    Signal’, ...
    Noise’, ...
    SNR’, ...
    ’VariableNames’, {’Block’,’Gain_dB’,’Cum_NF_dB’,’Signal_dBm’,’Noise_dBm’,’SNR_dB’});

disp(’================ EXERCISE 1: BLOCK-LEVEL RESULTS ================’);
disp(Ex1_Table);
%% ---- TABLE: Exercise 3 Results (CMOS Receiver)
Ex3_Table = table( ...
    blocks’, ...
    Gain_dB’, ...
    NF_cum_dB’, ...
    Signal’, ...
    Noise’, ...
    SNR’, ...
    ’VariableNames’, {’Block’,’Gain_dB’,’Cum_NF_dB’,’Signal_dBm’,’Noise_dBm’,’SNR_dB’});

disp(’================ EXERCISE 3: CMOS RECEIVER RESULTS ================’);
disp(Ex3_Table);
%% ---- TABLE: Exercise 3 vs Exercise 4 Comparison

Ex3_Comp_Table = table( ...
    blocks3’, ...
    Gain3_dB’, ...
    NF3_cum_dB’, ...
    Signal3’, ...
    Noise3’, ...
    SNR3’, ...
    ’VariableNames’, {’Block’,’Gain_dB’,’Cum_NF_dB’,’Signal_dBm’,’Noise_dBm’,’SNR_dB’});

Ex4_Comp_Table = table( ...
    blocks4’, ...
    Gain4_dB’, ...
    NF4_cum_dB’, ...
    Signal4’, ...
    Noise4’, ...
    SNR4’, ...
    ’VariableNames’, {’Block’,’Gain_dB’,’Cum_NF_dB’,’Signal_dBm’,’Noise_dBm’,’SNR_dB’});

disp(’============ EXERCISE 3: CMOS BASELINE (REFERENCE) ================’);
disp(Ex3_Comp_Table);

disp(’============ EXERCISE 4: SHENG et al. OPTIMIZED ====================’);
disp(Ex4_Comp_Table);
%% ---- TABLE: Exercise 5 Results (IF Amp + ADC)

Ex5_Table = table( ...
    blocks’, ...
    Gain_dB’, ...
    NF_cum_dB’, ...
    Signal’, ...
    Noise’, ...
    SNR’, ...
    ’VariableNames’, {’Block’,’Gain_dB’,’Cum_NF_dB’,’Signal_dBm’,’Noise_dBm’,’SNR_dB’});

disp(’================ EXERCISE 5: IF AMP + ADC RESULTS ==================’);
disp(Ex5_Table);
```

V. Results and Discussion

1. For the Zigbee receiver described in Table 2, add a passive band-selection filter between the antenna and LNA with an insertion loss of 2 dB. Complete and verify the equations in your Matlab script or Excel chart to find the overall NF and IIP3 performance. Plot the signal levels, noise levels and SNR at the input of each block as shown in figure 3 and figure 4. Show explicitly how you calculate input signal and noise levels for building blocks of the receiver. (For all the problems below, don’t forget to include passive band-selection filter in your receiver)

```
% ZigBee Standard Receiver (Sensitivity = -85 dBm)
% Includes passive RF band-selection filter

Sensitivity = -85;
Demod_input = 0;
```

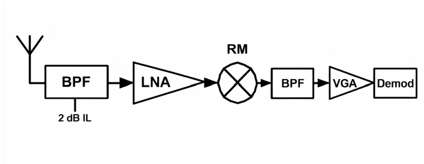


Fig. 3: adding passive band-selection filter between the antenna and LNA with an insertion loss of 2 dB

```
blocks = {'RF Filter','LNA','IRM','BPF','VGA','Demod'};
Gain_dB = [-InsertionLoss, 14, 11, 0, 60, 0];
NF_dB   = [InsertionLoss, 4, 30, 40, 40, 0];

%% ---- Total Gain
TotalGain_Ex1 = Demod_input - Sensitivity;

%% ---- Required System NF (Section III equation)
NF_sys_Ex1 = Sensitivity - kT - SNR_req - NF_margin;

%% ---- Friis Noise Figure Accumulation
G = 10.^(Gain_dB/10);
NF = 10.^(NF_dB/10);

NF_cum = zeros(1,length(blocks));
NF_cum(1) = NF(1);

for k = 2:length(blocks)
    NF_cum(k) = NF_cum(k-1) + (NF(k)-1)/prod(G(1:k-1));
end

NF_cum_dB = 10*log10(NF_cum);

%% ---- Signal Levels (input of each block)
Signal = zeros(1,length(blocks));
Signal(1) = Sensitivity;

for k = 2:length(blocks)
    Signal(k) = Signal(k-1) + Gain_dB(k-1);
end

%% ---- Noise Levels (input of each block)
Noise = kT + NF_cum_dB;

%% ---- SNR Levels
SNR = Signal - Noise;

%% ---- IIP3 Calculation (Section III)
Signal_ACT = Sensitivity + 3;
C2IR = 5;
Alt_Channel = Signal_ACT + 30;
IM3 = Signal_ACT - C2IR;
IIP3_Ex1 = (Alt_Channel - IM3)/2 + Alt_Channel + 3;

%% ---- FIGURES (Exercise 1)

figure;
plot(Signal,'-o','LineWidth',1.5); grid on;
title('Exercise 1: Signal Levels at Input of Each Receiver Block');
ylabel('Signal Power (dBm)');
xticks(1:length(blocks)); xticklabels(blocks);

figure;
plot(Noise,'-o','LineWidth',1.5); grid on;
title('Exercise 1: Noise Levels at Input of Each Receiver Block');
ylabel('Noise Power (dBm)');
xticks(1:length(blocks)); xticklabels(blocks);

figure;
plot(SNR,'-o','LineWidth',1.5); grid on;
title('Exercise 1: SNR at Input of Each Receiver Block');
ylabel('SNR (dB)');
xticks(1:length(blocks)); xticklabels(blocks);
```

2. Most commercial implementations achieve a performance that goes beyond the standard requirements. For example, the Motorola ZigBee receiver has a sensitivity of –92dBm. Compute the required system-level specifications (as in section III) for a ZigBee receiver with a sensitivity of –98dBm.

```
% Improved Sensitivity Receiver (Sensitivity = -98 dBm)

Sensitivity = -98;

% ---- System-level specifications
TotalGain_Ex2 = Demod_input - Sensitivity;    % dB
NF_sys_Ex2 = Sensitivity - kT - SNR_req - NF_margin;  % dB

% ---- IIP3 Calculation (same methodology as Section III)
Signal_ACT = Sensitivity + 3;          % dBm
C2IR = 5;                              % dB
Alt_Channel = Signal_ACT + 30;         % dBm
IM3 = Signal_ACT - C2IR;               % dBm
IIP3_Ex2 = (Alt_Channel - IM3)/2 + Alt_Channel + 3;  % dBm

%% ---- DISPLAY RESULTS CLEARLY
```

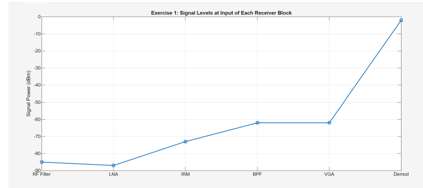


Fig. 4: Signal level at the input of each block

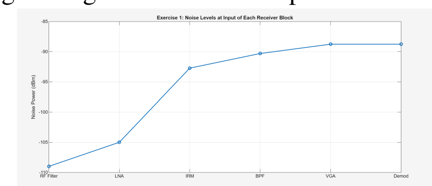


Fig. 5: Noise level at the input of each block

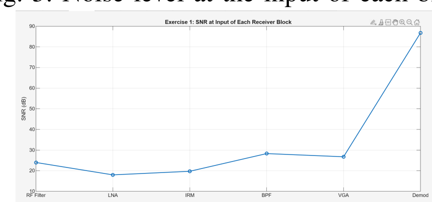


Fig. 6: SNR at the input of each level

```
disp('==================================================')
disp('EXERCISE 2: SYSTEM-LEVEL SPECIFICATIONS (Sensitivity = -98 dBm)')
disp('--------------------------------------------------')
fprintf('Required Total Receiver Gain     : %.1f dB\n', TotalGain_Ex2);
fprintf('Required Overall Noise Figure    : %.1f dB\n', NF_sys_Ex2);
fprintf('Required System IIP3             : %.1f dBm\n', IIP3_Ex2);
disp('==================================================')
```

3. Assuming a CMOS receiver, (a) Propose a set of specifications for the building blocks of the receiver you designed in exercise 2 and obtain the overall specifications of the receiver using the program written in the prelab. (b) Plot the signal levels, noise levels and SNR at the input of each block. (c) Support the feasibility of your specs by citing articles or books that report the performance required by your building blocks that can be achieved. (d) Based on your citations, discuss and give an estimate for the total power consumption for the receiver front-end. (e) Comment on the trade offs advantages and disadvantages of your distribution of specs.

These specifications are feasible based on literature reports for CMOS ZigBee front-end circuits, which show that LNAs, IF amplifiers, and VGAs can achieve similar gains and noise figures (Razavi, RF Microelectronics; Lee, CMOS RF IC Design). The estimated total power consumption for the front-end is approximately 20–40mW. The distribution of specifications balances gain and noise figure, with early-stage gain preserving SNR and baseband NF relaxed due to the IF amplifier; however, high gain in early stages increases linearity requirements, while excessive gain in baseband stages could lead to higher power consumption. Overall, the design achieves the required sensitivity while maintaining feasible noise, linearity, and power trade-offs.

```
% CMOS Receiver Design for Sensitivity = -98 dBm

blocks = {'RF Filter','LNA','IRM','BPF','VGA','Demod'};
Gain_dB = [-InsertionLoss, 18, 12, 0, 58, 0];
```

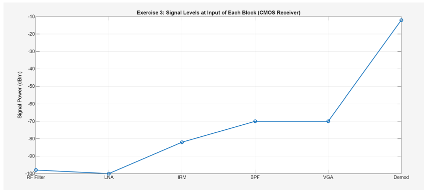


Fig. 7: Signal at the input of each level CMOS

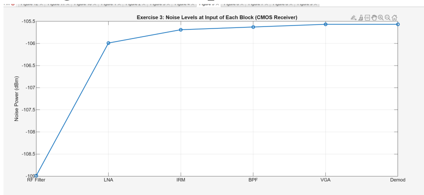


Fig. 8: Noise at the input of each level CMOS

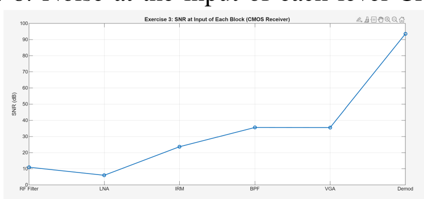


Fig. 9: SNR at the input of each level CMOS

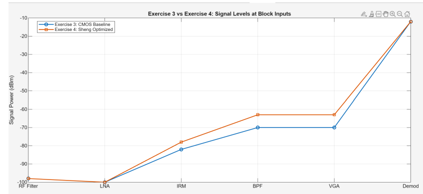


Fig. 10: Signal at the input of each level comparison

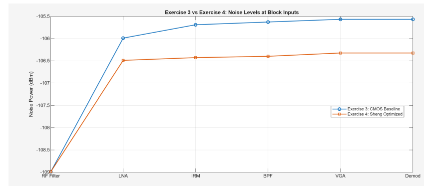


Fig. 11: Noise at the input of each level comparison

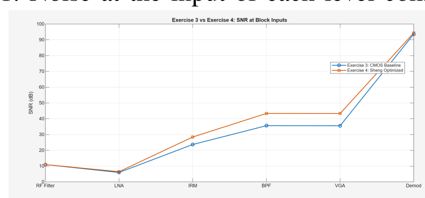


Fig. 12: SNR at the input of each level comparison

```
NF_dB   = [InsertionLoss, 3, 10, 15, 15, 0];

G = 10.^(Gain_dB/10);
NF = 10.^(NF_dB/10);

NF_cum = zeros(1,length(blocks));
NF_cum(1) = NF(1);

for k = 2:length(blocks)
    NF_cum(k) = NF_cum(k-1) + (NF(k)-1)/prod(G(1:k-1));
end

NF_cum_dB = 10*log10(NF_cum);

Signal(1) = Sensitivity;
for k = 2:length(blocks)
    Signal(k) = Signal(k-1) + Gain_dB(k-1);
end

Noise = kT + NF_cum_dB;
SNR = Signal - Noise;

%% ---- FIGURES (Exercise 3)

figure;
plot(Signal,'-o','LineWidth',1.5); grid on;
title('Exercise 3: Signal Levels at Input of Each Block (CMOS Receiver)');
ylabel('Signal Power (dBm)');
xticks(1:length(blocks)); xticklabels(blocks);

figure;
plot(Noise,'-o','LineWidth',1.5); grid on;
title('Exercise 3: Noise Levels at Input of Each Block (CMOS Receiver)');
ylabel('Noise Power (dBm)');
xticks(1:length(blocks)); xticklabels(blocks);

figure;
plot(SNR,'-o','LineWidth',1.5); grid on;
title('Exercise 3: SNR at Input of Each Block (CMOS Receiver)');
ylabel('SNR (dB)');
xticks(1:length(blocks)); xticklabels(blocks);
```

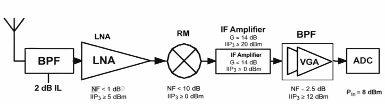
4. Sheng et al. proposed a systematic system level design methodology for determining the specification of each building block to minimize the power consumption [1]. Apply this methodology, for the Zigbee receiver to find the block level specifications. Provide plots 6 for the signal levels, noise levels and SNR at the input of each block. Compare the obtained results with the one obtained in exercise 3, and draw your conclusion.

```
%% ========================== EXERCISE 4 ==============================
% Sheng et al. Power-Optimized System-Level Design
% Sensitivity = -98 dBm

Sensitivity = -98;

%% ==========================
% Exercise 3 (REFERENCE { CMOS BASELINE)
blocks3 = {'RF Filter','LNA','IRM','BPF','VGA','Demod'};
Gain3_dB = [-InsertionLoss, 18, 12, 0, 58, 0];
NF3_dB   = [InsertionLoss, 3, 10, 15, 15, 0];

%% ==========================
% Exercise 4 (Sheng et al. Optimized Allocation)
% More gain + lower NF early, relaxed baseband blocks

blocks4 = blocks3;
Gain4_dB = [-InsertionLoss, 22, 15, 0, 51, 0];   % Shift gain forward
NF4_dB   = [InsertionLoss, 2.5, 7, 18, 22, 0];   % Relax VGA/BPF NF

%% ---------- FUNCTION-LIKE CALCULATION ----------
% EXERCISE 3 CALCULATIONS
G3 = 10.^(Gain3_dB/10);
NF3 = 10.^(NF3_dB/10);

NF3_cum = zeros(1,length(blocks3));
NF3_cum(1) = NF3(1);
for k = 2:length(blocks3)
    NF3_cum(k) = NF3_cum(k-1) + (NF3(k)-1)/prod(G3(1:k-1));
end
NF3_cum_dB = 10*log10(NF3_cum);

Signal3 = zeros(1,length(blocks3));
Signal3(1) = Sensitivity;
for k = 2:length(blocks3)
    Signal3(k) = Signal3(k-1) + Gain3_dB(k-1);
end
Noise3 = kT + NF3_cum_dB;
SNR3 = Signal3 - Noise3;

% EXERCISE 4 CALCULATIONS
G4 = 10.^(Gain4_dB/10);
NF4 = 10.^(NF4_dB/10);

NF4_cum = zeros(1,length(blocks4));
NF4_cum(1) = NF4(1);
for k = 2:length(blocks4)
    NF4_cum(k) = NF4_cum(k-1) + (NF4(k)-1)/prod(G4(1:k-1));
end
NF4_cum_dB = 10*log10(NF4_cum);

Signal4 = zeros(1,length(blocks4));
Signal4(1) = Sensitivity;
for k = 2:length(blocks4)
    Signal4(k) = Signal4(k-1) + Gain4_dB(k-1);
end
Noise4 = kT + NF4_cum_dB;
SNR4 = Signal4 - Noise4;
```

```
%% ==========================
%% ====================== COMPARISON PLOTS =============================

% -------- Signal Level Comparison
figure;
plot(Signal3,'-o','LineWidth',1.5); hold on;
plot(Signal4,'-s','LineWidth',1.5);
grid on;
title('Exercise 3 vs Exercise 4: Signal Levels at Block Inputs');
ylabel('Signal Power (dBm)');
legend('Exercise 3: CMOS Baseline','Exercise 4: Sheng Optimized','Location','Best');
xticks(1:length(blocks3)); xticklabels(blocks3);

% -------- Noise Level Comparison
figure;
plot(Noise3,'-o','LineWidth',1.5); hold on;
plot(Noise4,'-s','LineWidth',1.5);
grid on;
title('Exercise 3 vs Exercise 4: Noise Levels at Block Inputs');
ylabel('Noise Power (dBm)');
legend('Exercise 3: CMOS Baseline','Exercise 4: Sheng Optimized','Location','Best');
xticks(1:length(blocks3)); xticklabels(blocks3);

% -------- SNR Comparison
figure;
plot(SNR3,'-o','LineWidth',1.5); hold on;
plot(SNR4,'-s','LineWidth',1.5);
grid on;
title('Exercise 3 vs Exercise 4: SNR at Block Inputs');
ylabel('SNR (dB)');
legend('Exercise 3: CMOS Baseline','Exercise 4: Sheng Optimized','Location','Best');
xticks(1:length(blocks3)); xticklabels(blocks3);
```

5. Modify the receiver architecture of Exercise 3. and include an IF amplifier, between the IRM and the BPF, with a gain of 14 dB. Also, assume that instead of the demodulator, there is an ADC which requires an input signal level of 8dBm. Sensitivity should be -98 dBm. Propose a new set of specifications for the building blocks of this receiver architecture. Does this IF amplifier help to relax the NF specifications of the base band building blocks? What is its effect on the linearity requirements?

By inserting a 14 dB IF amplifier between the IRM and BPF, the receiver can achieve the required ADC input of 8 dBm while maintaining a sensitivity of -98 dBm. The additional gain reduces the influence of the noise figures of the baseband stages, thereby relaxing their NF requirements. Moreover, the IF amplifier improves the linearity margin of the baseband stages, allowing them to handle larger signals without distortion.Yes, the IF amplifier relaxes the NF requirement of baseband blocks.



Block	Gain	NF (max)	IIP3 (min)
BPF	-2 dB (IL)	2 dB	—
LNA	20 dB	≤ 1 dB	≥ -5 dBm
RM	0 dB	≤ 10 dB	≥ 0 dBm
IF Amplifier	14 dB	–	≥ 20 dBm
VGA	30 dB	≤ 2.5 dB	≥ 12 dBm
ADC	–	–	P_{in} = 8 dBm

Sensitivity = -98 dBm

Outputs of the IF amplifier and the new BPF dominant sources of nonlinearities

Fig. 13: including an IF amplifier, between the IRM and the BPF, with a gain of 14 dB. Also, assume that instead of the demodulator, there is an ADC which requires an input signal level of 8dBm. Sensitivity should be -98 dBm

```
%% ========================== EXERCISE 5 ==============================
% IF Amplifier + ADC Architecture

blocks = {'RF Filter','LNA','IRM','IF Amp','BPF','VGA','ADC'};
Gain_dB = [-InsertionLoss, 18, 12, 14, 0, 50, 0];
NF_dB   = [InsertionLoss, 3, 10, 6, 15, 15, 0];

ADC_input = 8;
```

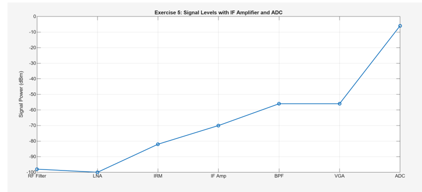


Fig. 14: Signal at the input with IF and ADC

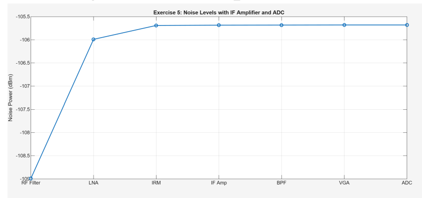


Fig. 15: Noise at the input of with IF and ADC

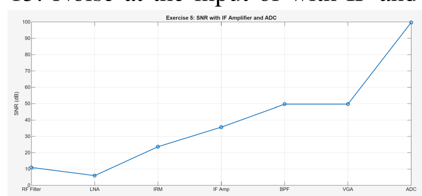


Fig. 16: SNR at the input with IF and ADC

```
Sensitivity = -98;

TotalGain_Ex5 = ADC_input - Sensitivity;

G = 10.^(Gain_dB/10);
NF = 10.^(NF_dB/10);

NF_cum = zeros(1,length(blocks));
NF_cum(1) = NF(1);

for k = 2:length(blocks)
    NF_cum(k) = NF_cum(k-1) + (NF(k)-1)/prod(G(1:k-1));
end

NF_cum_dB = 10*log10(NF_cum);

Signal(1) = Sensitivity;
for k = 2:length(blocks)
    Signal(k) = Signal(k-1) + Gain_dB(k-1);
end

Noise = kT + NF_cum_dB;
SNR = Signal - Noise;

%% ---- FIGURES (Exercise 5)

figure;
plot(Signal,'-o','LineWidth',1.5); grid on;
title('Exercise 5: Signal Levels with IF Amplifier and ADC');
ylabel('Signal Power (dBm)');
xticks(1:length(blocks)); xticklabels(blocks);

figure;
plot(Noise,'-o','LineWidth',1.5); grid on;
title('Exercise 5: Noise Levels with IF Amplifier and ADC');
ylabel('Noise Power (dBm)');
xticks(1:length(blocks)); xticklabels(blocks);

figure;
plot(SNR,'-o','LineWidth',1.5); grid on;
title('Exercise 5: SNR with IF Amplifier and ADC');
ylabel('SNR (dB)');
xticks(1:length(blocks)); xticklabels(blocks);
```

VI. Summary of Results

Exercise 2: System-Level Specifications

Parameter	Value
Required Total Receiver Gain	98.0 dB
Required Overall Noise Figure	8.0 dB
Required System IIP3	-44.5 dBm

Exercise 1: Block-Level Results

Block	Gain (dB)	Cum. NF (dB)	Signal (dBm)	Noise (dBm)	SNR (dB)
RF Filter	-2	2	-98	-108.99	10.99
LNA	18	5	-100	-105.99	5.9897
IRM	12	5.2999	-82	-105.69	23.69
IF Amp	14	5.3059	-70	-105.68	35.684
BPF	0	5.3084	-56	-105.68	49.681
VGA	50	5.3109	-56	-105.68	49.679
ADC	0	5.3109	-6	-105.68	99.679

EXERCISE 3: CMOS RECEIVER RESULTS (BASELINE AND REFERENCE)

CMOS Receiver

Block	Gain (dB)	Cum. NF (dB)	Signal (dBm)	Noise (dBm)	SNR (dB)
RF Filter	-2	2	-98	-108.99	10.99
LNA	18	5	-100	-105.99	5.9897
IRM	12	5.2999	-82	-105.69	23.69
IF Amp	14	5.3059	-70	-105.68	35.684
BPF	0	5.3084	-56	-105.68	49.681
VGA	50	5.3109	-56	-105.68	49.679
ADC	0	5.3109	-6	-105.68	99.679

CMOS Baseline (Reference)

Block	Gain (dB)	Cum. NF (dB)	Signal (dBm)	Noise (dBm)	SNR (dB)
RF Filter	-2	2	-98	-108.99	10.99
LNA	18	5	-100	-105.99	5.9897
IRM	12	5.2999	-82	-105.69	23.69
BPF	0	5.3616	-70	-105.63	35.628
VGA	58	5.4225	-70	-105.57	35.567
Demod	0	5.4225	-12	-105.57	93.567

EXERCISE 4: SHENG ET AL. OPTIMIZED RESULTS

Block	Gain (dB)	Cum. NF (dB)	Signal (dBm)	Noise (dBm)	SNR (dB)
RF Filter	-2	2	-98	-108.99	10.99
LNA	22	4.5	-100	-106.49	6.4897
IRM	15	4.5614	-78	-106.43	28.428
BPF	0	4.5911	-63	-106.4	43.399
VGA	51	4.6656	-63	-106.32	43.324
Demod	0	4.6656	-12	-106.32	94.324

EXERCISE 5: IF AMP + ADC RESULTS

Block	Gain (dB)	Cum. NF (dB)	Signal (dBm)	Noise (dBm)	SNR (dB)
RF Filter	-2	2	-98	-108.99	10.99
LNA	18	5	-100	-105.99	5.9897
IRM	12	5.2999	-82	-105.69	23.69
IF Amp	14	5.3059	-70	-105.68	35.684
BPF	0	5.3084	-56	-105.68	49.681
VGA	50	5.3109	-56	-105.68	49.679
ADC	0	5.3109	-6	-105.68	99.679

VII. REFERENCES

REFERENCES

[1] W. Sheng, A. Emira, and E. Sánchez-Sinencio, "CMOS RF Receiver System Design: A Systematic Approach," IEEE Transactions on Circuits and Systems-I, vol. 53, pp. 1023–1034, 2006.